

UNITED STATES PATENT APPLICATION

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TITLE OF INVENTION

TECHNICAL FIELD

The present invention relates to structured training signal generation for generative artificial intelligence models, and more particularly to a compounding scenario engine that produces dynamically chained, music-industry-targeted training scenarios that propagate across training sessions via a persistent seed-chain mechanism, thereby teaching diffusion models the complete visual and narrative language of the professional music business.

BACKGROUND OF THE INVENTION

Diffusion models trained on generic image-text datasets learn broad visual competence but lack domain-specific fluency – the ability to generate imagery and conditioning prompts that accurately reflect the vocabulary, visual aesthetics, business logic, and narrative arcs of a specific professional domain. Fine-tuning on static curated datasets addresses this in part, but does not capture the temporal narrative structure of a professional domain: the fact that events cause other events, that career trajectories follow recognizable arcs, and that different roles within the same industry encounter systematically different types of challenges.

In the music industry specifically, a content creator encountering a viral post faces a predictably different set of subsequent events than a touring professional encountering a sold-out show. Prior art training systems treat every training step as independent: the model is presented with a scene-prompt pair without context about what preceded or follows it. This independence assumption discards the rich causal narrative structure of the training domain, resulting in models that can generate individual frames but cannot internalize the sequential, consequence-driven visual vocabulary of professional music careers.

There exists a need for a training signal generator that (1) selects training scenarios from a structured library organized by professional role and career stage, (2) produces consequence seeds that carry the causal logic of the domain forward into subsequent training steps, (3) tracks compound depth so that follow-up scenarios earn proportionally higher informational premiums, and (4) persists seed chains across training sessions to enable multi-session narrative arcs.

SUMMARY OF THE INVENTION

The present invention provides a compounding music industry scenario engine comprising a structured scenario library organized by job family and event type, a consequence seed propagation mechanism that carries narrative context across training steps, a career arc modeling component that weights career stages according to realistic industry distributions, and a state persistence layer that maintains active seeds across training sessions.

In a first aspect, the invention provides a system for generating structured training scenarios for a diffusion model comprising: a scenario library organized into a plurality of job families, each job family containing a plurality of event types, each event type specifying consequence seeds, an intensity range, applicable platforms, and visual conditioning templates; a seed queue that stores active consequence seeds propagated from previously fired scenarios; a career stage sampler that weights career stages according to a configurable distribution; a job family selector that uses inverse-exposure weighting optionally biased by gradient health signals; and a state serializer that persists the seed queue across process restarts.

In a second aspect, the invention provides a compound depth tracker that assigns increasing year-equivalent step premiums to scenarios drawn from deeper levels of a consequence chain,

reflecting the greater informational density of contextually situated domain knowledge versus randomly sampled prompts.

In a third aspect, the invention provides a scene category mapping that associates each job family with a set of valid visual scene categories, enabling thematic continuity between the training scenario narrative and the visual conditioning prompt fed to the diffusion model.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

I. Scenario Library Structure

The scenario library (SCENARIO_LIBRARY) is a three-level nested dictionary organized as:

```
job_family → event_type → {seeds, intensity, platforms,
                             visuals}
```

Eight job families are defined, each mapping to a MaxCore model target:

```
content_creator → content model
social_strategist → social model
ads_manager → advertising model
fan_engagement → engagement model
visual_director → UNetV4 visual generation model (primary)
release_architect → all four models simultaneously
touring_pro → content + social models
sync_composer → advertising + content models
```

Each event type contains four fields:

```
seeds: list of seed type strings that this event propagates
       to future steps
intensity: (low, high) tuple defining the uniform sampling
           range for scenario intensity
```

platforms: list of applicable social/distribution platforms for context

visuals: list of visual conditioning text templates for the diffusion model

In the preferred embodiment, the library contains at least 56 event types across 8 job families, with at least 4 visual conditioning templates per event type, yielding a minimum of 224 distinct visual conditioning strings.

II. Consequence Seed Mechanism

Each fired scenario produces a set of consequence seeds from its event type's seed list. Seeds are stored in an in-memory seed queue with the following fields per entry:

seed_type: string identifying the type of consequential event

chain_id: UUID identifying the narrative chain this seed belongs to

compound_depth: integer depth of this seed within the chain (0 = fresh, 1+ = follow-up)

weight: float weight = $1.0 + 0.5 \times \text{compound_depth}$ (deeper seeds are preferred)

planted_at: wall-clock timestamp for TTL enforcement

planted_step: training step number when the seed was planted

Seeds are claimed by sampling the queue weighted by the weight field. This ensures that deeper consequence chains are more likely to be continued than shallow or fresh chains, creating narrative momentum within the training signal.

A seed TTL (SEED_TTL_SECONDS = 1800, configurable) ensures that seeds planted in one session are not reused in sessions that occur more than 30 minutes later, preventing the propagation of stale narrative context.

The SEED_RESOLUTION table maps each seed_type to a (job_family, event_type) pair, enabling the engine to directly resolve a seed into a specific trainable scenario without requiring a full library search.

III. Compound Depth and YE-Step Premium

The compound_depth of a scenario tracks how deep within a consequence chain the current scenario sits:

- depth 0: fresh scenario, not the result of any seed
- depth 1: direct consequence of a depth-0 scenario
- depth 2+: deeper follow-up in an established narrative chain

Year-equivalent step credits are assigned as follows:

- depth 0: SCENARIO_YE_WEIGHT_BASE = 18 YE-steps (1.5× priority replay)
- depth 1: SCENARIO_YE_WEIGHT_COMPOUND_1 = 24 YE-steps (2× priority replay)
- depth 2+: SCENARIO_YE_WEIGHT_COMPOUND_2 = 30 YE-steps (2.5× priority replay)

This tiered premium reflects the observation that a contextually situated training example – one that logically follows from a preceding scenario – carries more informational density per gradient update than a randomly sampled scene, because it reinforces not only the visual content of the frame but also the causal business logic connecting events in the domain.

IV. Career Stage Modeling

The engine samples career stages from a discrete distribution reflecting the realistic population of artists at each stage of professional development:

- unsigned_indie: 12% (bedroom musician, no team, no budget)
- emerging: 25% (first traction, small loyal audience)
- breaking: 28% (fast-rising, industry attention, first deals)

established: 20% (proven chart/streaming success, touring income)

mainstream: 10% (household name, radio, stadium tours)

legacy: 5% (catalogue value, nostalgia tours)

Each career stage applies a visual modifier string to the conditioning prompt (`_STAGE_MODS`), shifting the aesthetic vocabulary from "underground indie raw bedroom authentic" at the unsigned level to "legendary iconic timeless classic historic" at the legacy level.

V. Job Family Selection

When no seed is available in the queue (or the 40% seed-claim probability roll fails), the engine selects a job family using inverse-exposure weighting:

$$\text{weight}(\text{family}) = 1.0 / (\text{exposure_count}(\text{family}) + 1)$$

This base weight is optionally biased by gradient health signals: if the gradient memory reports that scenes associated with a particular job family have vanishing or exploding gradient norms, the weight for that family is multiplied by $\max(0.2, 2.0 - \text{health_score})$, directing training attention toward underperforming visual domains.

VI. Scene Category Mapping and Visual Prompt Construction

Each job family maps to a set of valid scene categories (`SCENE_MAP`). When building a scenario, the engine selects a scene category from this set with a 35% probability of remaining in the current scene hint (if the hint is a valid scene for the chosen family) and a 65% probability of selecting a new scene uniformly from the family's valid set.

The visual conditioning prompt is constructed by selecting one of the event type's visual templates and appending career stage modifiers, intensity modifiers, and compound depth modifiers. Intensity modifiers range from "subtle intimate close warm" at

intensity ≤ 0.30 to "dramatic intense climactic cinematic" at intensity ≥ 0.80 . Compound depth ≥ 2 appends "layered compound evolving rich" to reflect the narrative depth of the training scenario.

VII. State Persistence

The engine serializes the following state fields to a JSON file at session end:

- active_seeds: the current seed queue
- job_exposure: per-family firing counts
- scene_exposure: per-scene firing counts
- total_fired: total scenarios fired across all sessions
- chain_counter: global chain ID counter

On initialization, the engine deserializes this state and removes expired seeds (older than SEED_TTL_SECONDS), enabling multi-session narrative chains that span process restarts.

VIII. Public API

The engine exposes four public methods:

- roll_scenario(scene_hint, gradient_health) $\rightarrow$ ScenarioSpec | None
- plant_seeds(spec, step_count) $\rightarrow$ None
- build_spec_for_family(job_family, scene_hint) $\rightarrow$ ScenarioSpec
- commit_spec(spec) $\rightarrow$ None
- save() / load()
- status() $\rightarrow$ dict

The ScenarioSpec dataclass contains: scenario_id, chain_id, job_family, model_target, career_stage, platform, event_type, intensity, compound_depth, consequence_seeds, scene_category, scene_prompt, ye_weight, and context_description. This structured output enables downstream training loop integration

without requiring the training loop to understand the scenario library format.

CLAIMS

1. A computer-implemented system for generating structured training scenarios for a generative diffusion model, the system comprising: a processor; and a non-transitory computer-readable medium storing instructions that, when executed by the processor, cause the processor to: maintain a scenario library organized into a plurality of professional domain job families, each job family containing a plurality of event types, each event type specifying at least one consequence seed type, an intensity range, and at least one visual conditioning template; maintain a seed queue storing active consequence seeds, each seed specifying a seed type, a narrative chain identifier, and a compound depth; select a training scenario by either claiming a seed from the seed queue or performing a fresh job family selection; construct a scenario specification comprising a job family, event type, career stage, intensity value, scene category, visual conditioning prompt, and year-equivalent step credit; and plant new consequence seeds from the fired scenario's seed list into the seed queue for use in subsequent training steps.
2. The system of claim 1, wherein each active seed in the seed queue carries a weight equal to one plus a configurable multiplier times the compound depth, causing seeds from deeper narrative chains to be selected with higher probability than seeds from shallower chains.
3. The system of claim 1, wherein the year-equivalent step credit assigned to a scenario specification increases monotonically with the compound depth of the scenario, reflecting the greater informational density of contextually situated domain training examples.

4. The system of claim 3, wherein the year-equivalent step credit at compound depth zero is a base value, the credit at compound depth one is between 1.3 and 1.5 times the base value, and the credit at compound depth two or greater is between 1.6 and 1.8 times the base value.
5. The system of claim 1, wherein the seed queue enforces a time-to-live constraint on each seed, removing seeds whose elapsed time since planting exceeds a configurable threshold, preventing stale narrative context from propagating across training sessions separated by a significant time interval.
6. The system of claim 1, further comprising a career stage sampler that selects a career stage from a plurality of career stages weighted according to a distribution reflecting the realistic population of professionals at each stage of the target domain.
7. The system of claim 1, wherein the fresh job family selection applies inverse-exposure weighting that assigns lower selection probability to job families that have been selected more frequently in prior training steps, ensuring balanced coverage of all job families over time.
8. The system of claim 7, wherein the inverse-exposure weighting is further biased by gradient health signals that increase the selection weight of job families associated with visual scene categories exhibiting vanishing or exploding gradient norms during recent training steps.
9. The system of claim 1, further comprising a scene category mapping that associates each job family with a set of valid visual scene categories, and wherein the visual conditioning prompt is constructed by combining a visual template from the event type with career stage modifiers

and intensity modifiers derived from the sampled intensity value.

10. The system of claim 1, further comprising a state serializer that persists the seed queue, per-family firing counts, per-scene firing counts, and a global chain identifier counter to a persistent storage medium at the end of each training session, and restores said state at the start of a subsequent training session.
11. The system of claim 1, wherein each scenario specification further specifies a model target field that identifies which one or more component models of a multi-model training infrastructure should receive the gradient update generated from the scenario.
12. The system of claim 11, wherein at least one job family maps to a model target of all available component models simultaneously, and wherein said job family earns a higher year-equivalent step credit premium than job families targeting a single component model.
13. A method of generating compounding training signals for a domain-specific generative model, the method comprising: firing a first training scenario from a structured scenario library, the first scenario belonging to a first professional domain job family and event type; extracting consequence seed types from the fired scenario and storing them in a persistent seed queue with a compound depth value incremented from the fired scenario's depth; firing a second training scenario by claiming one of the stored consequence seeds, wherein the second scenario's job family and event type are resolved from the claimed seed type; assigning a higher year-equivalent training credit to the second scenario than to the first scenario, based on the increased compound depth of the second scenario; and continuing the chain by

extracting consequence seeds from the second scenario and storing them with a further incremented compound depth.

14. The method of claim 13, further comprising persisting the seed queue to non-volatile storage between training sessions, enabling narrative chains to span multiple training runs separated by process restarts.
15. The method of claim 13, wherein the compound depth is bounded by a maximum value beyond which all scenarios receive the same maximum year-equivalent credit.
16. The method of claim 13, further comprising associating each scenario with a career stage selected from a distribution of career stages weighted to reflect the realistic population distribution of professionals in the target domain.
17. A non-transitory computer-readable medium storing instructions that, when executed by a processor, implement a scenario specification structure comprising: a job family field identifying a professional role category within a target domain; an event type field identifying a specific business event within the job family; a compound depth field recording the depth of the scenario within a consequence chain; a consequence seeds field listing seed types that should be propagated to future training scenarios; a scene category field identifying the visual environment associated with the scenario; a scene prompt field containing a ready-to-use visual conditioning string for a diffusion model; a year-equivalent weight field specifying the informational density credit for the training step; and a chain identifier field linking the scenario to its narrative chain for multi-session tracking.
18. The computer-readable medium of claim 17, further storing instructions that implement a seed resolution table

mapping each consequence seed type to a target job family and event type pair, enabling direct resolution of a seed into a specific trainable scenario without library search.

- 19.** The computer-readable medium of claim 17, further storing instructions that implement a build-spec-for-family method that generates a scenario specification for a specified job family without modifying any global firing count, session counter, or last-scenario reference, enabling a plurality of variant specifications to be generated speculatively and evaluated before any single specification is committed to the training state.
- 20.** The computer-readable medium of claim 17, further storing instructions that implement a commit-spec method that, when called with a single selected scenario specification, updates global firing counts, scene exposure counts, total fired count, and last-scenario reference atomically, ensuring that only one scenario per training step is recorded as having influenced the training state regardless of how many speculative variants were generated.

ABSTRACT

A compounding music industry scenario engine generates structured training scenarios for diffusion model fine-tuning by maintaining a hierarchical scenario library organized into eight professional job families, each containing a plurality of event types with consequence seeds, intensity ranges, platform associations, and visual conditioning templates. When a scenario is fired, its consequence seeds are planted into a persistent seed queue with escalating weights and compound depth values. Subsequent training steps preferentially claim these seeds, continuing narrative chains that teach the model the causal business logic of the music industry. Scenarios at greater compound depth earn proportionally higher year-equivalent training credits, reflecting the greater informational density of contextually situated domain knowledge. Career stages are sampled from a distribution reflecting the realistic population of music professionals. The seed queue is serialized across training sessions, enabling multi-session narrative arcs. A build-spec-for-family method enables speculative multi-variant generation for upstream quality gate integration without corrupting global training state.